Universität Bonn

Notes for the lecture

Combinatorics, Graphs, Matroids

held by

Dr. Ulrich Brenner

TEXed by

Daniel Emse

Winter 2024/25

Corrections and improvements

If you have corrections or improvements, contact me via (`daniel.emse@uni-bonn.de`).

Contents

Lecture **1**

1 Combinatorics **2**

Appendix **A**

List of definitions **B**

List of statements **C**

Index **D**

Lecture

Chapter 1

Combinatorics

[08.10.2024, Lecture 1]

Goal: Count the number of elements in some finite sets.

Lemma 1.1: Fundamental counting rules

(*i*) *For pairwise disjoint and finite sets* $U_1, \dots, U_n$*, we have*

$$\left| \bigcup_{i=1}^{n} U_i \right| = \sum_{i=1}^{n} |U_i| . \tag{1.1}$$

(*ii*) *For finite sets* $U_1, \dots, U_n$*, we have*

$$|U_1 \times \cdots \times U_n| = \prod_{i=1}^{n} |U_i| . \tag{1.2}$$

(*iii*) *If there is a bijection between two sets* S *and* T*, then* $|S| = |T|$*.*

Notation. We call a set a n-set if it is a set with cardinality $n \in \mathbb{N}$ (where $0 \in \mathbb{N}$).

Proposition 1.2. *For two* n*-sets* A *and* B*, the number of bijections between* A *and* B *is* $n!$*.*

Proof. Let U the set of bijections from A to B. Apply induction in n:

$n = 0$: Trivial, because $0! = 1 = |U|$.

$n \mapsto n + 1$: Let $n > 0$ and let $x \in A$. For $i \in B$, let U_i be the set of all bijections $f : A \to B$ with $f(x) = i$. By induction, $|U_i| = (n-1)!$ for each $i \in B$. Then, $U = \bigcup_{i \in B} U_i$, so we get

$$|U| = \sum_{i \in B} |U_i| = n \cdot (n-1)! = n!. \tag{1.3}$$

□

Corollary 1.3. *The number of permutations of an* n*-set is* $n!$*.* □

Proposition 1.4. *The number of mappings from a* k*-set* A *to an* n*-set* B *is* n^k*.*

Proof. There is a bijection between the set of all mappings from A to B and the set $\underbrace{B \times \cdots \times B}_{k \text{ times}}$.

The number of these mappings is by Lemma 1.1.(*ii*) just $|B|^k = n^k$. □

Notation. We denote the set of all mappings from A to B by B^A.

Proposition 1.5. *The number of subsets of an* n*-set is* 2^n*.*

Proof. For an n-set A, there is a bijection between its power set and the mappings from A to $\{0,1\}$. Thus, the power set has 2^n elements. □

Notation.

(*i*) We denote the power set of a set A by 2^A.

(*ii*) The *binomial coefficient* $\binom{n}{k}$ is defined as the number of k-subsets of a n-set (for $k, n \in \mathbb{N}$).

What does *counting* mean? ⇝ Possible answers:

(*i*) a direct, closed formula,

(*ii*) a sum,

(*iii*) a recursive formula

Example. Consider the number y_n of $0-1-2$-strings of length n with an even number of 1s and an odd number of 2s. Answers of types (*ii*) and (*iii*):

- We have
$$y_n = \sum_{i=0}^{\lfloor \frac{n-1}{2} \rfloor} \binom{2}{2i+1} 2^{2i}. \tag{1.4}$$
- We have
$$y_n = 3^{n-1} - y_{n-1} \quad (n > 0). \tag{1.5}$$

The proof of both formulas is exercise 1.2.

Proposition 1.6. *For $n, k \in \mathbb{N}$ with $k \le n$, we have*

(*a*) $\binom{n}{k} = \binom{n}{n-k}$

(*b*) $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$ *for* $k \ge 1$

(*c*) $\binom{n}{k} = \frac{n!}{(n-k)!k!}$.

Consider three kinds of rows in Pascal's triangle:

Row sums $\sum_{k=0}^{n} \binom{n}{k} = 2^n$ (both terms count the number of subsets of an n-set)

Column sums $\sum_{m=0}^{n} \binom{m}{k} = \binom{n+1}{k+1}$ for $n, k \in \mathbb{N}$ (where $\binom{m}{k}$ is the number of all $(k+1)$-subsets of $\{1, \dots, n+1\}$ with the property that $m+1$ is the largest element in the subset)

Diagonal sums $\sum_{k=0}^{n} \binom{m+k}{k} = \binom{m+n+1}{n}$ for $m, n \in \mathbb{N}$ (for $k \in \{0, \dots, n\}$, where $\binom{m+n}{k}$ is the number of n-subsets of $\{1, \dots, m+n+1\}$ where $m+k+1$ is the largest number that is **not** contained in the subset)

Proposition 1.7. *For $x, y \in \mathbb{R}$, we have $(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}$.*

Proof. The coefficients of $x^k y^{n-k}$ in $(x+y)^n$ is $\binom{n}{k}$. □

Definition 1.8: Stirling number

For $n, k \in \mathbb{N}$ let $S_{n,k}$ be the number of ways to partition an n-set in to k non-empty sets. The number $S_{n,k}$ is called *Stirling number of the second kind.*

⇝ for $n > 0 : S_{n,0} = 0, S_{n,1} = n, S_{n,2} = 2^{n-1} - 1$.

Proposition 1.9. *For $n, k \in \mathbb{N} \setminus \{0\}$ we have $S_{n,k} = S_{n-1,k-1} + k \cdot S_{n-1,k}$.*

Proof. Let M be an n-set and $x \in M$. Then, the set of partitions of n into k subsets can be decomposed in the set of partitions with the set $\{x\}$ (there are $S_{n-1,k-1}$ of them) and the set of partitions where x is an element in a set with more than one element (there are $k \cdot S_{n-1,k}$ of them). □

[08.10.2024, Lecture 1]
[10.10.2024, Lecture 2]

Appendix

List of definitions

1.8 Stirling number 3

List of statements

1.1 Fundamental counting rules 2

Index

binomial coefficient, 3

Stirling number, 3
    - of second kind, 3